

MATHEMATICS

TARGET : JEE- Advanced

CAPS-6**MATRICES & DETERMINANTS****SCQ (Single Correct Type) :**

1. Let $a = \lim_{x \rightarrow 1} \frac{x}{\ln x} - \frac{1}{x \ln x}$; $b = \lim_{x \rightarrow 0} \frac{x^3 - 16x}{4x + x^2}$; $c = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x)}{x}$ and $d = \lim_{x \rightarrow -1} \frac{(x+1)^3}{3(\sin(x+1) - (x+1))}$, then the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is
 (A) Idempotent (B) Involutory (C) Non singular (D) Nilpotent
2. A is a 2×2 matrix such that $A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $A^2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Find the sum of the elements of A.
 (A) -1 (B) 0 (C) 2 (D) 5
3. Matrix A satisfies $A^2 = 2A - I$ where I is the identity matrix then for $n \geq 2$, A^n is equal to ($n \in \mathbb{N}$)
 (A) $nA - I$ (B) $2^{n-1}A - (n-1)I$ (C) $nA - (n-1)I$ (D) $2^{n-1}A - I$
4. If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & a & 1 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & c \\ 5/2 & -3/2 & 1/2 \end{bmatrix}$, then
 (A) $a = 1, c = -1$ (B) $a = 2, c = -\frac{1}{2}$ (C) $a = -1, c = 1$ (D) $a = \frac{1}{2}, c = \frac{1}{2}$
5. The number of solution of the matrix equation $X^2 = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$ is
 (A) more than 2 (B) 2 (C) 1 (D) 0
6. 'A' is a 3×3 matrix with entries from the set $\{-1, 0, 1\}$. The probability that 'A' is neither symmetric nor skew symmetric is
 (A) $\frac{3^9 - 3^6 - 3^3 + 1}{3^9}$ (B) $\frac{3^9 - 3^6 - 3^3}{3^9}$ (C) $\frac{3^9 - 1}{3^{10}}$ (D) $\frac{3^9 - 3^3 + 1}{3^9}$
7. $A = \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$ and $MA = A^{2m}$, $m \in \mathbb{N}$ for some matrix M, then which one of the following is correct?
 (A) $M = \begin{bmatrix} a^{2m} & b^{2m} \\ b^{2m} & -a^{2m} \end{bmatrix}$ (B) $M = (a^2 + b^2)^m \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$
 (C) $M = (a^m + b^m) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (D) $M = (a^2 + b^2)^{m-1} \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$



8. Let 'A' and 'B' be two given matrices such that $AB = A$ and $BA = B$, then $A^2 B^2$ equals
- (A) A (B) B (C) I (D) 0

MCQ (One or more than one correct) :

9. P is a non-singular matrix and A, B are two matrices such that $B = P^{-1}AP$ then the true statements among the following are
- (A) A is invertible iff B is invertible
 (B) $B^n = P^{-1}A^nP \forall n \in \mathbb{N}$
 (C) $\forall \lambda \in \mathbb{R}, B - \lambda I = P^{-1}(A - \lambda I)P$ (I is the identity matrix)
 (D) A, B are both singular matrices
10. If A and B are respectively a symmetric and a skew symmetric matrix such that $AB = BA$ then
- (A) $(A - B)^{-1}(A + B)$ is orthogonal matrix when $(A - B)$ is non-singular
 (B) $(A + B)^{-1}(A - B)$ is orthogonal matrix when $(A + B)$ is non-singular
 (C) $\det[(A - B)^{-1}(A + B)] = 1$ and $\det[(A + B)^{-1}(A - B)] = -1$
 (D) $\det[(A - B)^{-1}(A + B)] = -1$ and $\det[(A + B)^{-1}(A - B)] = 1$
11. If $A = \begin{bmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{bmatrix}$ (where $\alpha_2 \neq \beta_1$ and $\alpha_1, \alpha_2, \beta_1, \beta_2$ are non-zero) satisfies the equation $x^2 + k = 0$, then
- (A) trace A = 0 (B) $\alpha_1 \beta_2 < 0$ (C) $\det A = k$ (D) $\det A = -k$
12. Let A is a $n \times n$ matrix in which diagonal elements are 1, 2, 3, ..., n (i.e., $a_{11} = 1, a_{22} = 2, a_{33} = 3, \dots, a_{ii} = i, \dots, a_{nn} = n$) and all other elements are equal to 'n' then
- (A) A_n is singular for all 'n' (B) A_n is non-singular for all 'n'
 (C) $\det A_5 = 120$ (D) $\det A_n = 0$
13. If the matrix A and B are of 3×3 and $(I - AB)^{-1}A$ is invertible, then which of the following statements is/are correct?
- (A) $I - BA$ is not invertible
 (B) $I - BA$ is invertible
 (C) $I - BA$ has for its inverse $I + B(I - AB)^{-1}A$
 (D) $I - BA$ has for its inverse $I + A(I - BA)^{-1}B$
14. If A, B are two square matrices of same order such that $A + B = AB$ and I is identity matrix of order same as that of A, B, then
- (A) $AB = BA$ (B) $|A - I| = 0$ (C) $|B - I| \neq 0$ (D) $|A - B| = 0$

Comprehension Type Question:**Omprehension # 1**

A Pythagorean triple is triplet of positive integers (a, b, c) such that $a^2 + b^2 = c^2$. Define the matrices A, B and C by

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -2 \\ 2 & 2 & 3 \end{bmatrix} \text{ and } C = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$$

15. If we write Pythagorean triples (a, b, c) in matrix form as [a, b, c] then which of the following matrix product is not a Pythagorean triplet?
- (A) [3,4,5]A (B) [3,4,5]B (C) [3,4,5]C (D) None of these
16. Which one of the following does not hold good?
- (A) $A^{-1} = \text{adj. } A$ (B) $(AB)^{-1} = \text{adj. } (AB)$
 (C) $(BC)^{-1} = \text{adj. } (BC)$ (D) $(ABC)^{-1} \neq \text{adj. } (ABC)$
17. $\text{Tr} (A + B^T + 3C)$ equals
- (A) 17 (B) 15 (C) 19 (D) 18

Omprehension # 2

Consider the matrix function

$$A(x) = \begin{bmatrix} \cos^{-1} x & \sin^{-1} x & \text{cosec}^{-1} x \\ \sin^{-1} x & \sec^{-1} x & \tan^{-1} x \\ \text{cosec}^{-1} x & \tan^{-1} x & \cot^{-1} x \end{bmatrix}$$

and $B = A^{-1}$. Also, $\det.(A(x))$ denotes the determinant of square matrix $A(x)$.

18. Which of the following statement(s) is(are) correct?
- (A) $A(-x) = A(x)$. (B) $A(x) + A(-x) = \pi I_3$
 (C) $A(-x) = -A(x)$. (D) $A(x) + A(-x) = -\pi I_3$
- [Note : I_3 denotes an identity matrix of order 3.]
19. Which of the following statement(s) is (are) correct?
- (A) $A(x)$ is a symmetric matrix. (B) $A(x)$ is a skew symmetric matrix.
 (C) Maximum value of $\det.(A(x))$ equals $\frac{\pi^3}{8}$ (D) Minimum value of $\det.(A(x))$ equals $\frac{\pi^3}{16}$.
20. Which of the following statement(s) is(are) correct?
- (A) $\det.(A(x))$ is a continuous function in its domain but not differentiable in its domain.
 (B) $\det.(A(x))$ is a continuous and differentiable function in its domain.
 (C) $\det.(A(x))$ is a bounded function.
 (D) $\det.(A(x))$ is one-one and odd function.

21. Which of the following statement(s) is(are) correct?

(A) If $a = \det. (B) + \det. (B^2) + \det. (B^3) + \dots \infty$, then minimum value of a equals $\frac{8}{\pi^3 - 8}$.

(B) If $b = \det. \text{adj.} (B) + \det. \text{adj.} (B^2) + \det. \text{adj.} (B^3) + \dots \infty$, then maximum value of b is $\frac{256}{\pi^6 - 256}$.

(C) If $a = \det. (B) + \det. (B^2) + \det. (B^3) + \dots \infty$, then maximum value of a equals $\frac{16}{\pi^3 - 16}$.

(D) If $b = \det. \text{adj.} (B) + \det. \text{adj.} (B^2) + \det. \text{adj.} (B^3) + \dots \infty$, then minimum value of b is $\frac{64}{\pi^6 - 64}$.

Numerical based Questions :

22. If $(\alpha\beta)(\delta\gamma) = \gamma\gamma\gamma$ such that $\alpha, \beta, \delta, \gamma$ represents a number from 1 to 9 & $\alpha, \beta, \delta, \gamma$ are all different digits & $\alpha\beta, \delta\gamma$ are two digit numbers & $\gamma\gamma\gamma$ is a three digit number, $\alpha > \delta$, and the trace of the matrix

$$A = \begin{bmatrix} \alpha & 1 & 2 & 0 \\ 0 & \beta & 1 & 1 \\ 0 & 0 & \gamma & 3 \\ 1 & 1 & 0 & \gamma \end{bmatrix} \text{ is a, then } \frac{a}{7} \text{ is equal to}$$

23. Let A be the set of all 3×3 symmetric matrices all of whose entries either 0 or 1. Five of these entries are 1 and four of them are 0. If n is the number of such matrices, then $n/2$ is

Matrix Match Type :

24. Match the following:

For 3×3 matrix, a_{ij} represents the elements of i^{th} row and j^{th} column.

Column-I		Column-II	
A	$a_{ij} + a_{jk} + a_{ki} = 0$, then	p	This represents a symmetric matrix
B	$a_{ij} - a_{jk} - a_{ki} = 0$, then	q	This represents a skew symmetric matrix
C	$(-1)^{i+j} a_{ij} + (-1)^{j+k} a_{jk} + (-1)^{k+i} a_{ki} = 0$, then	r	This represents a matrix whose all diagonal elements are zeros
		s	Value of determinant is zero

Code :

(A) $A \rightarrow (q, r, s); B \rightarrow (p, q, r, s); C \rightarrow (q, r, s)$

(B) $A \rightarrow (q, s); B \rightarrow (p, q, r, s); C \rightarrow (q, r, s)$

(C) $A \rightarrow (q, r, s); B \rightarrow (p, q, s); C \rightarrow (q, r, s)$

(D) $A \rightarrow (q, r, s); B \rightarrow (p, q, r, s); C \rightarrow (q, s)$

25. Match the following:

Column-I		Column-II	
A	If $ A = 2$ then $ 2A^{-1} =$ (where A is a matrix of order 3)	p	1
B	If $ A = \frac{1}{8}$ then $ \text{adj}(\text{adj } 2A) =$ (where A is matrix of order 3)	q	4
C	If $(A + B)^2 = A^2 + B^2$ and $ A = 2$ & $ B =$ (where A and B are matrices of odd order)	r	Does not exist
D	$ A_{2 \times 2} = 2$, $ B_{3 \times 3} = 3$ and $ C_{4 \times 4} = 4$ & $ ABC =$	s	0

Code:

(A) $A \rightarrow q$; $B \rightarrow p$; $C \rightarrow s$; $D \rightarrow r$ (B) $A \rightarrow q$; $B \rightarrow r$; $C \rightarrow p$; $D \rightarrow s$ (C) $A \rightarrow q$; $B \rightarrow r$; $C \rightarrow r$; $D \rightarrow s$ (D) $A \rightarrow q$; $B \rightarrow q$; $C \rightarrow s$; $D \rightarrow r$

ANSWER KEY OF CAPS-6

1. (D)	2. (D)	3. (C)	4. (A)	5. (A)
6. (A)	7. (D)	8. (A)	9. (ABC)	10. (AB)
11. (ABC)	12. (BC)	13. (BC)	14. (AC)	15. (A)
16. (D)	17. (A)	18. (B)	19. (ACD)	20. (AC)
21. (ABCD)	22. (4.00)	23. (6.00)	24. (A)	25. (A)